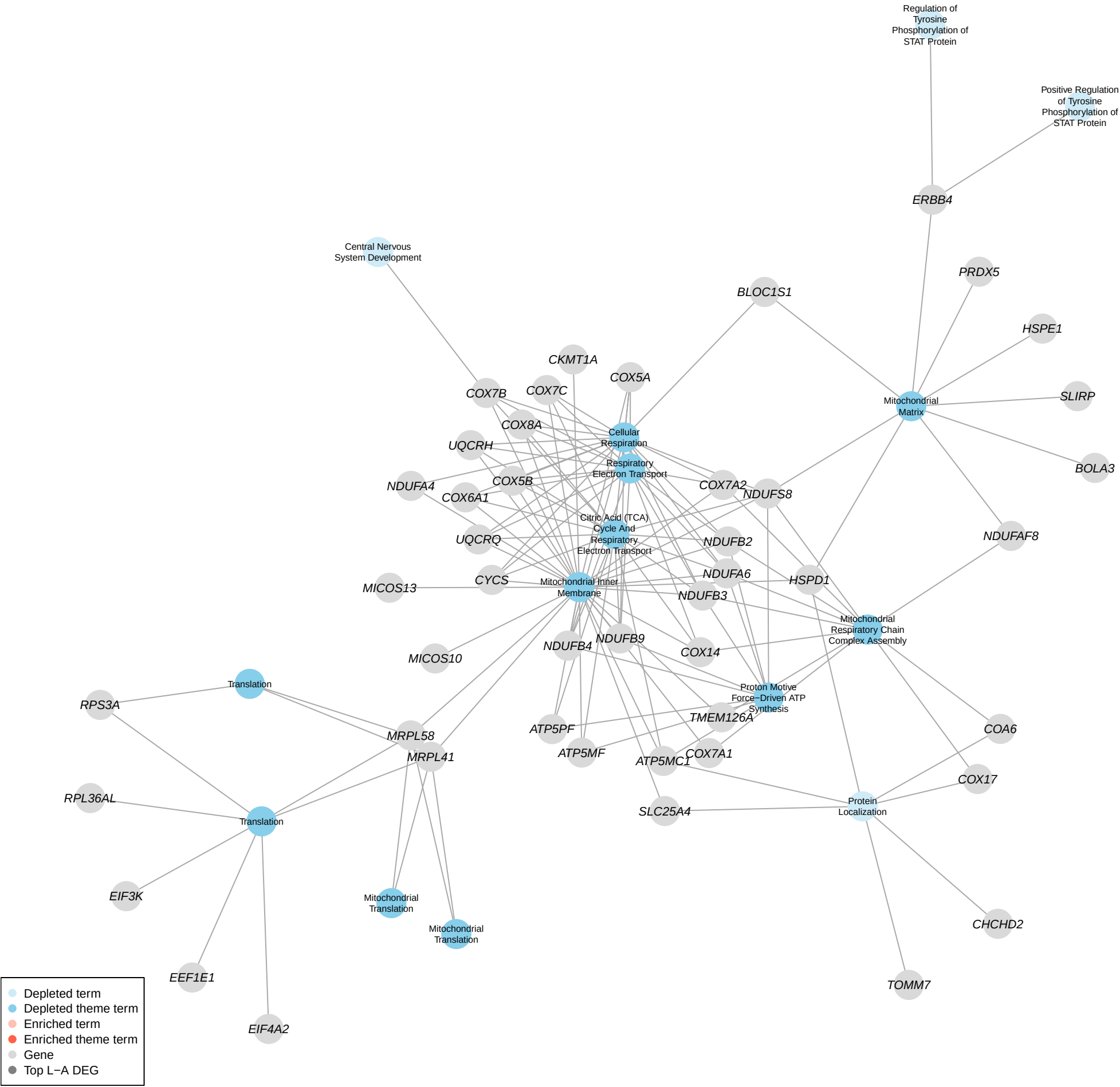
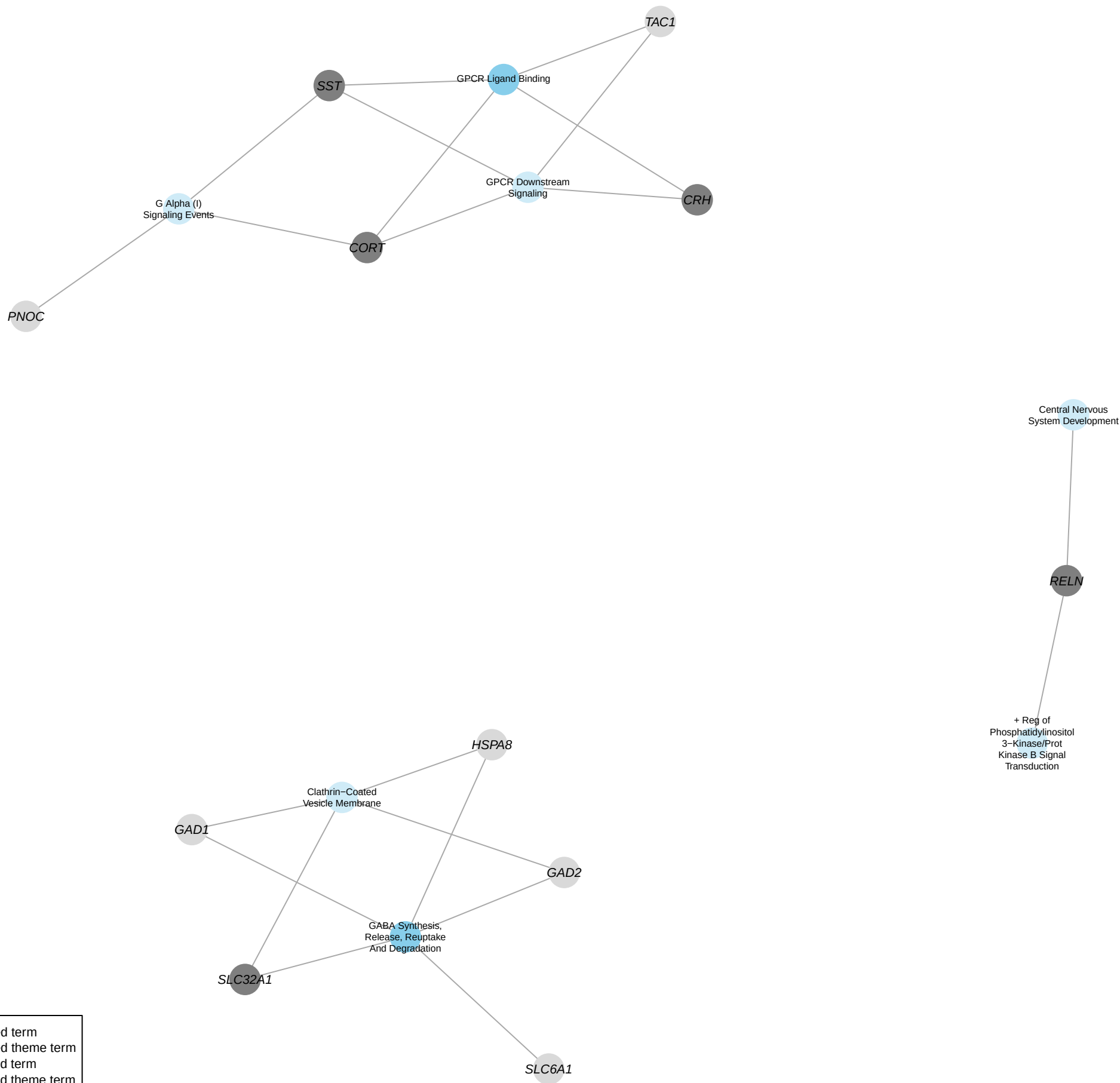


NTC.BPD M upkeep theme

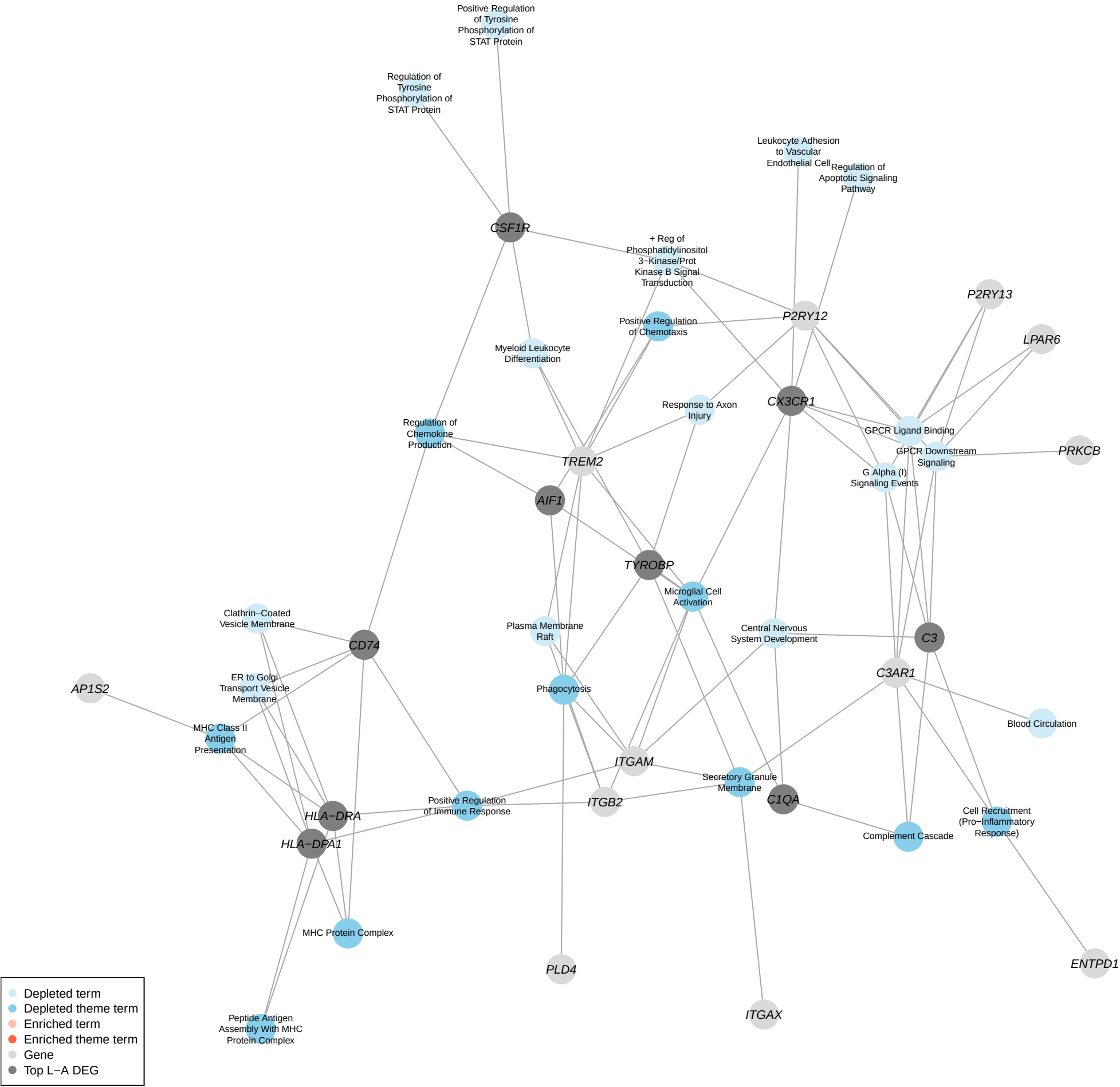


NTC.BPD M gaba theme

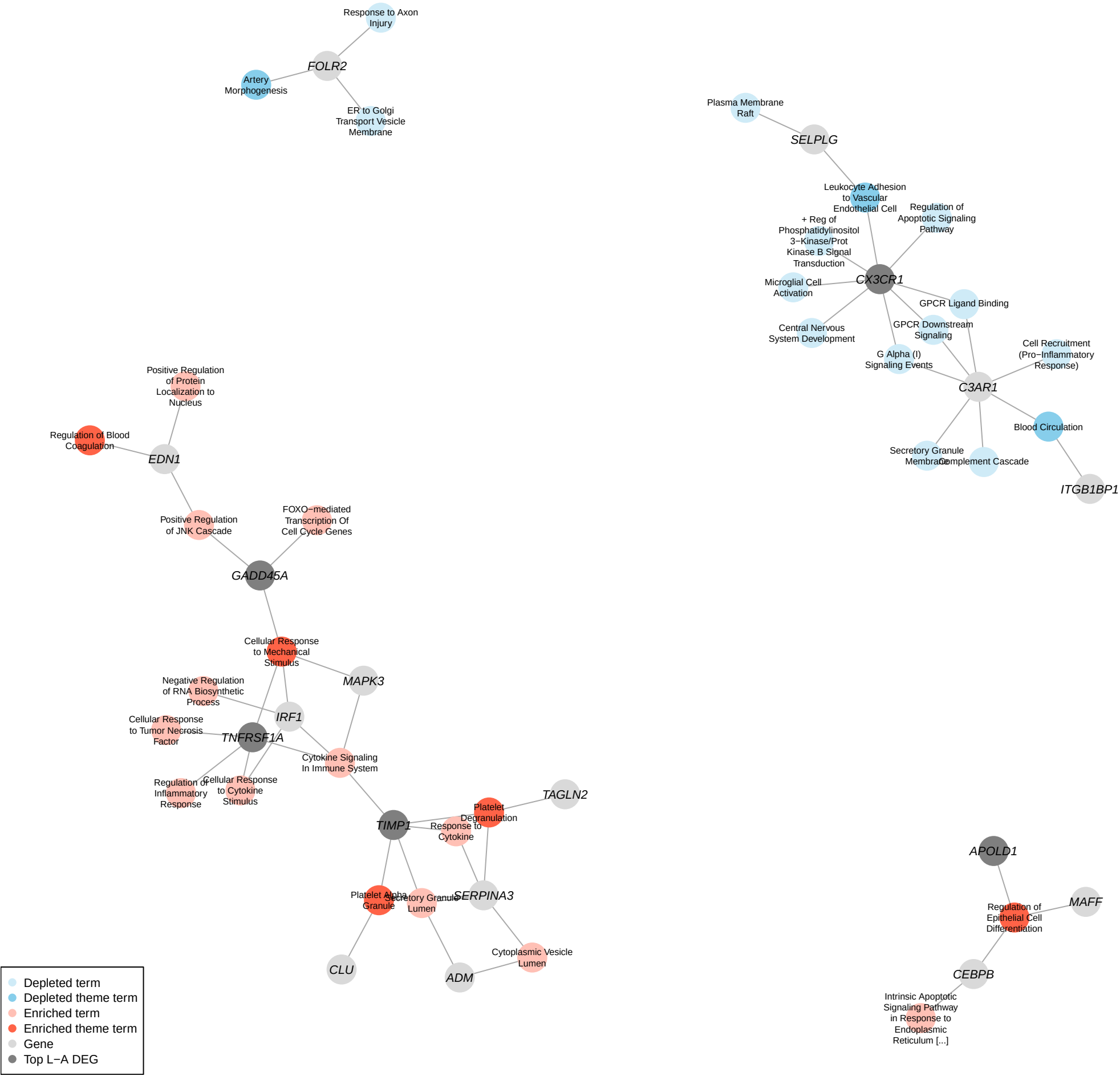


- Depleted term
- Depleted theme term
- Enriched term
- Enriched theme term
- Gene
- Top L-A DEG

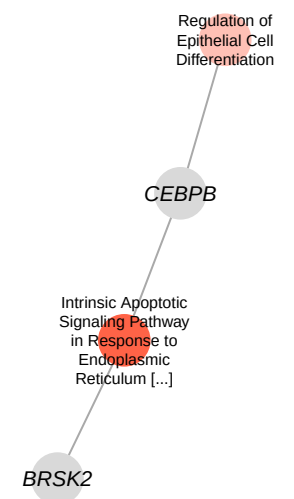
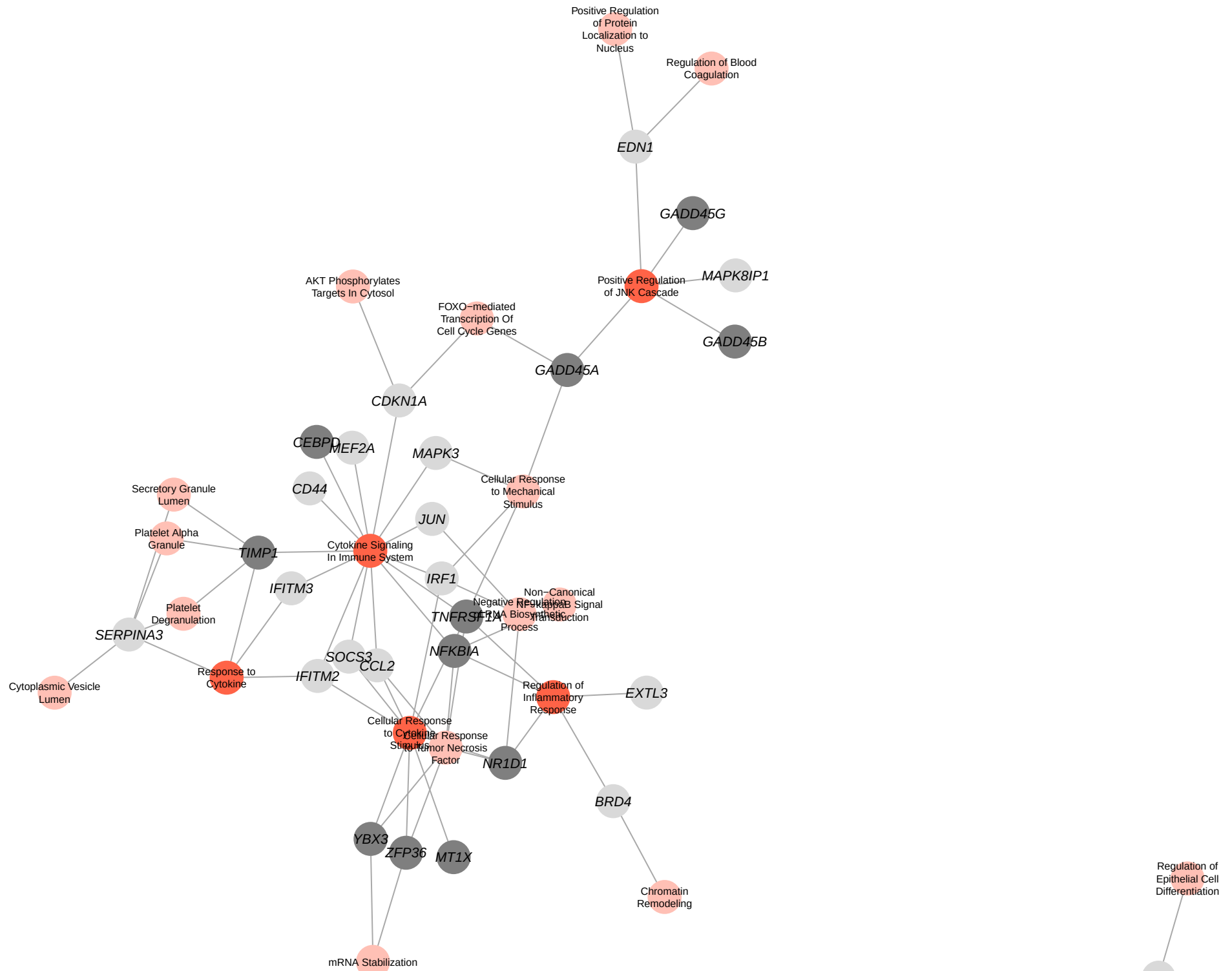
NTC.BPD M immune theme



NTC.BPD M vascular theme



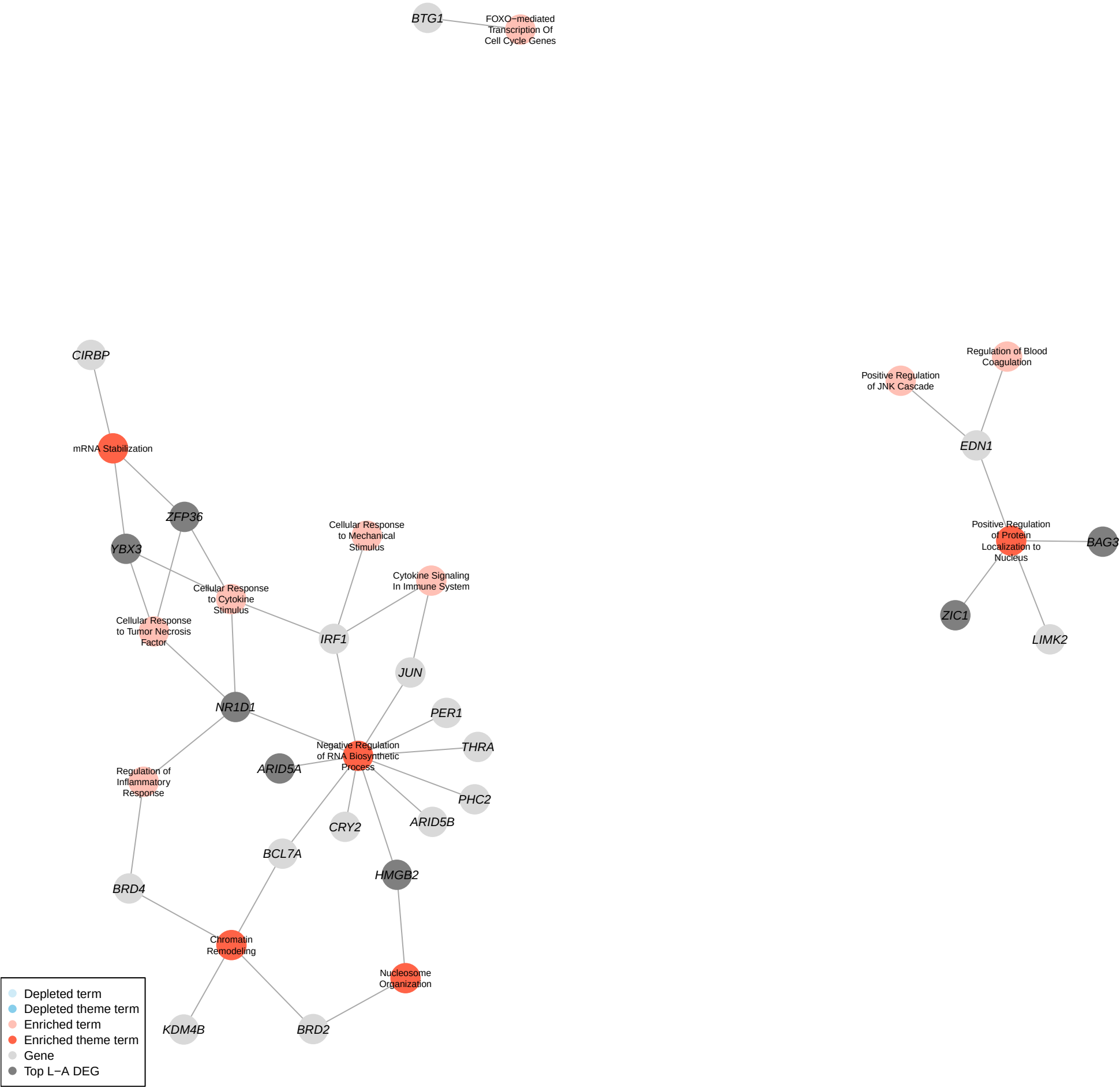
NTC.BPD M proinfl theme



- Depleted term
- Depleted theme term
- Enriched term
- Enriched theme term
- Gene
- Top L-A DEG

Reactome, GO-BP, and GO-CC results combined to produce theme

NTC.BPD M nuclear theme



Reactome, GO-BP, and GO-CC results combined to produce theme

Intersection Size

50
40
30
20
10
0

46

19

15

15

11

8

6

5

2

2



gaba



vascular



immune



nuclear



proinfl



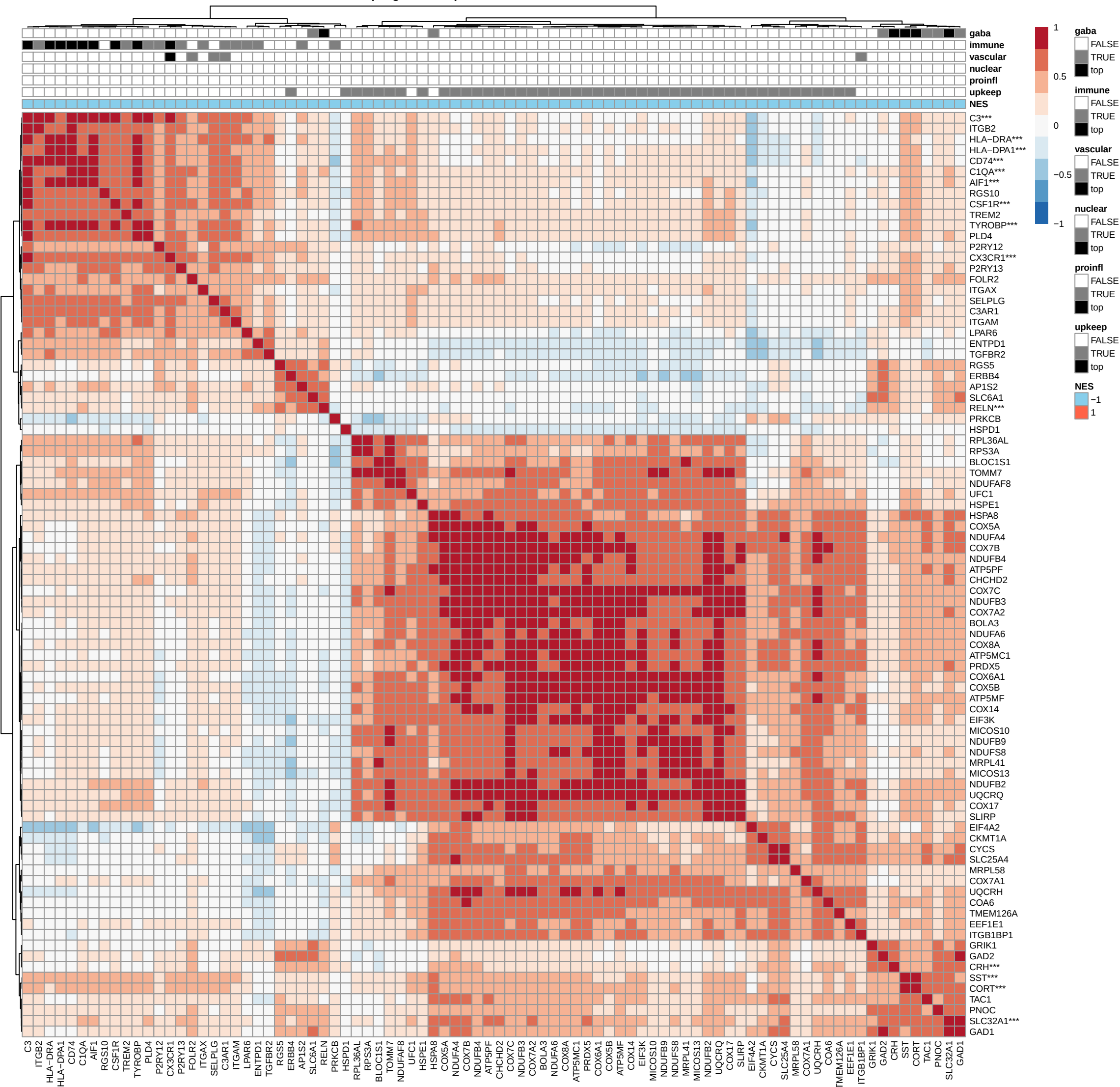
upkeep



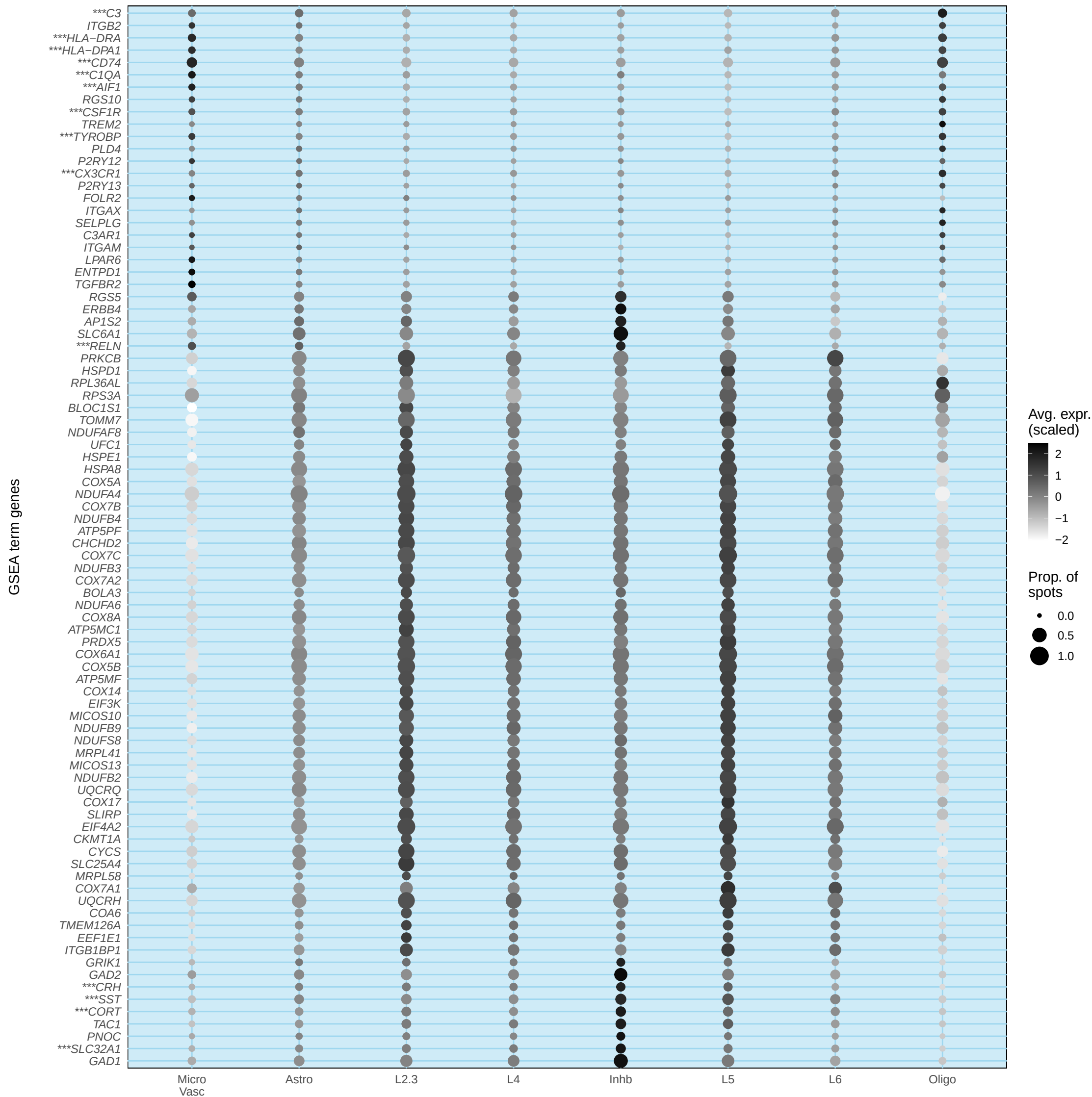
40 30 20 10 0

Set Size

NTC.BPD M L-A depleted genes
Top logFC DEGs per theme annotated with black



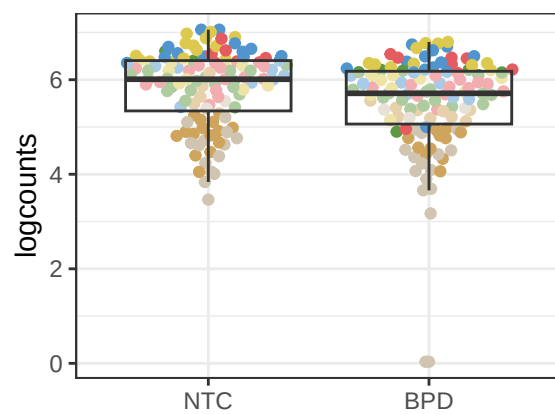
NTC.BPD M L-A depleted terms (**top L-A DEGs per theme)



CKMT1A

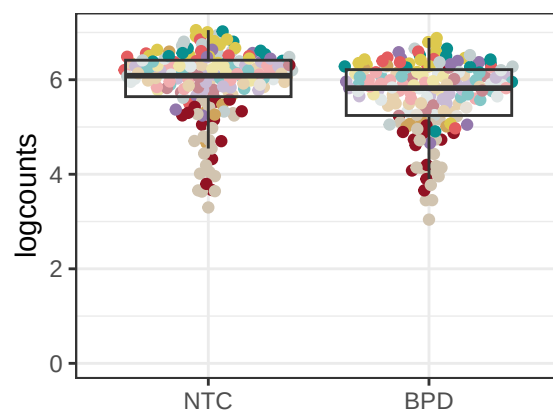
PRECAST (smoothed)

L-A sig: adj p< 5e-02 logFC = -0.27



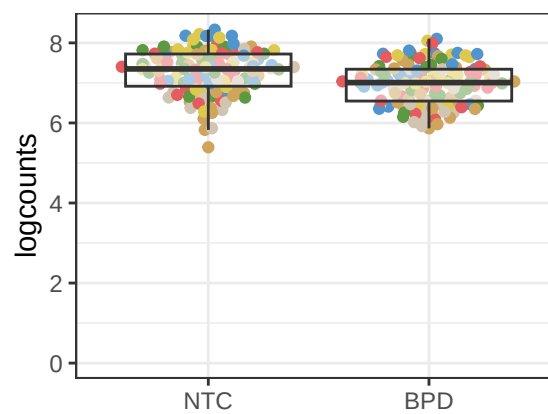
Seurat labels

L-A sig: adj p< 3e-02 logFC = -0.28

**NDUFB3**

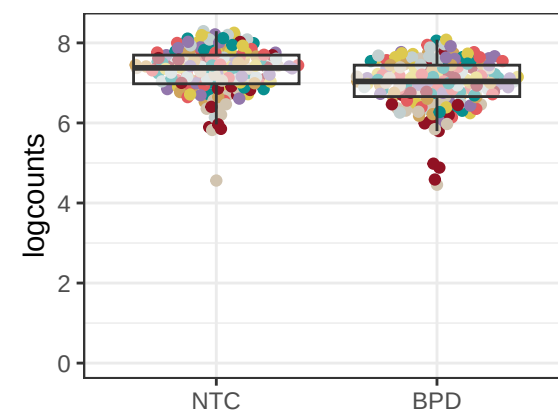
PRECAST (smoothed)

L-A sig: adj p< 1e-02 logFC = -0.26



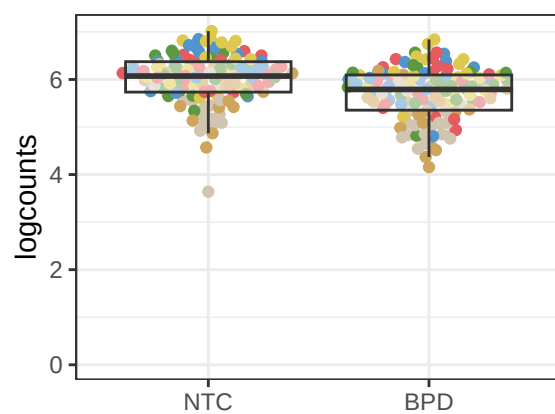
Seurat labels

L-A sig: adj p< 6e-03 logFC = -0.25

**COA6**

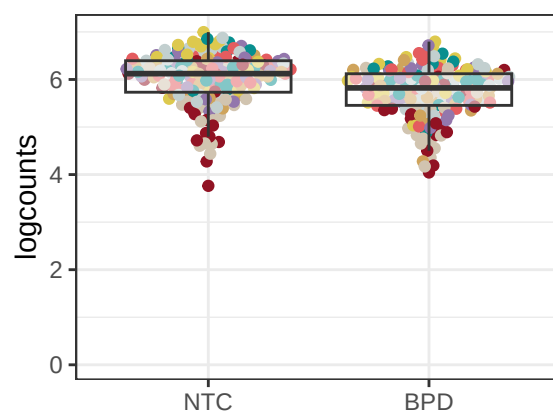
PRECAST (smoothed)

L-A sig: adj p< 2e-02 logFC = -0.25



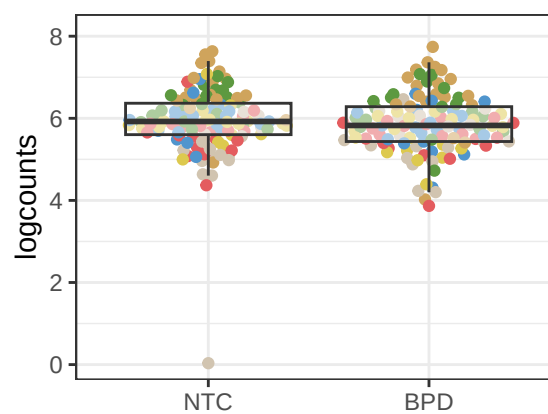
Seurat labels

L-A sig: adj p< 9e-03 logFC = -0.28

**ERBB4**

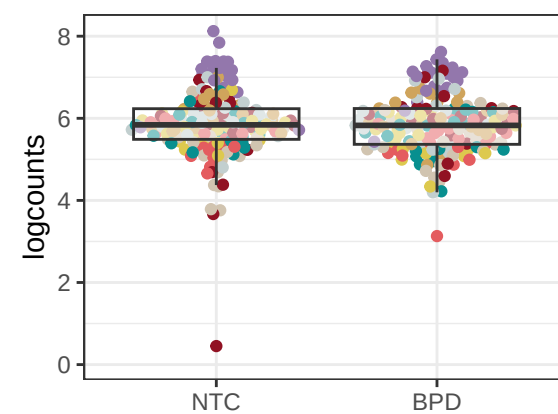
PRECAST (smoothed)

L-A sig: adj p< 5e-02 logFC = -0.23



Seurat labels

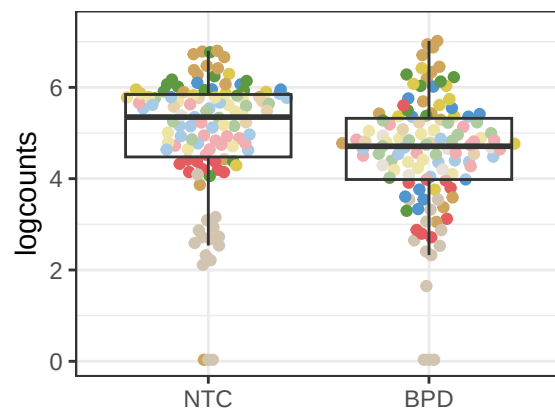
L-A sig: adj p< 2e-02 logFC = -0.25



CRH

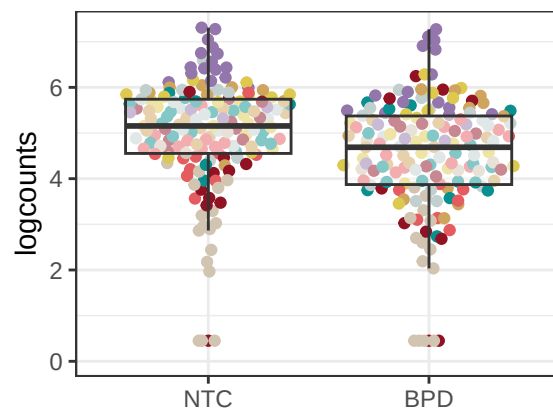
PRECAST (smoothed)

L-A sig: adj p< 2e-04 logFC = -0.61



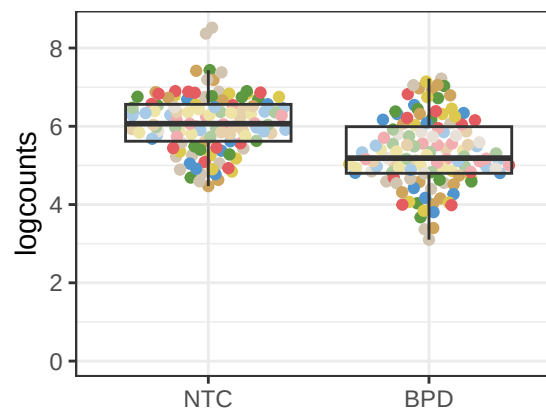
Seurat labels

L-A sig: adj p< 1e-03 logFC = -0.56

**CORT**

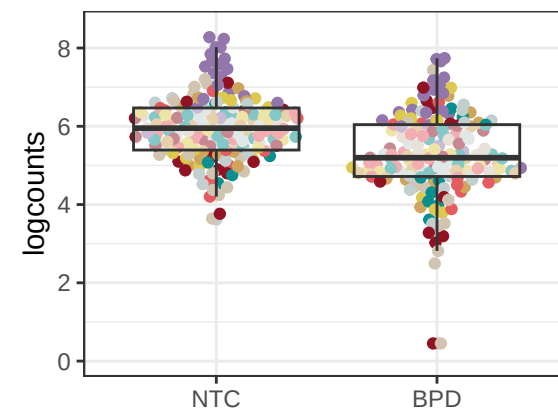
PRECAST (smoothed)

L-A sig: adj p< 2e-04 logFC = -0.61



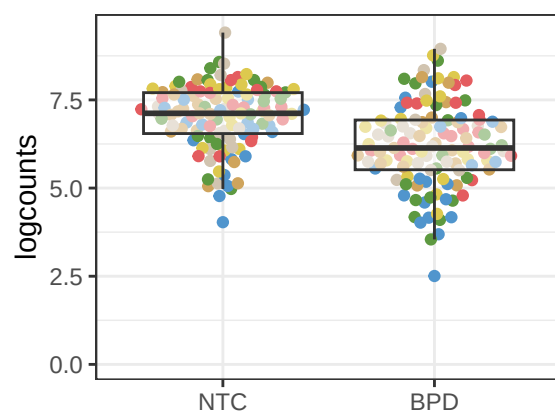
Seurat labels

L-A sig: adj p< 2e-04 logFC = -0.57

**SST**

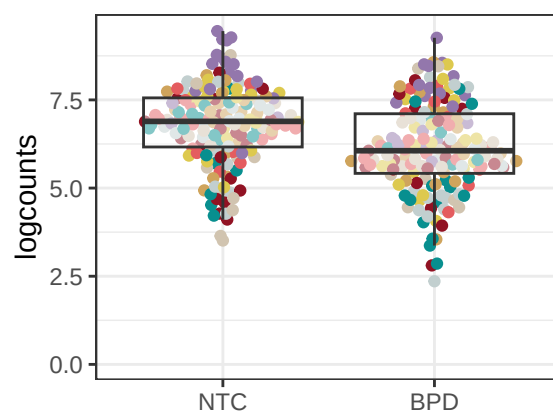
PRECAST (smoothed)

L-A sig: adj p< 2e-03 logFC = -0.81



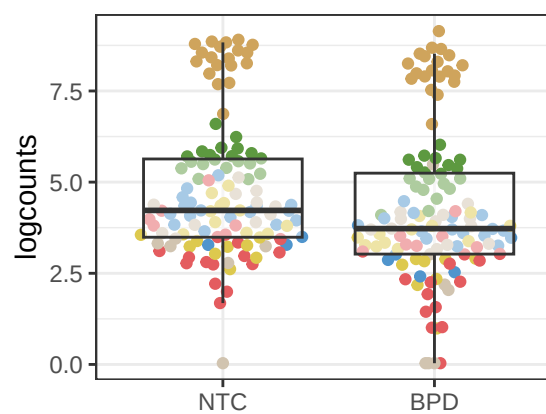
Seurat labels

L-A sig: adj p< 1e-03 logFC = -0.75

**RELN**

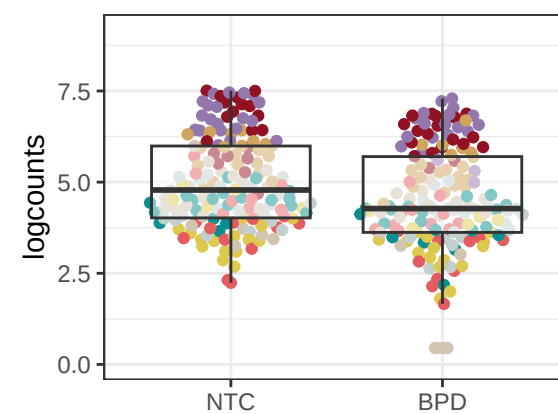
PRECAST (smoothed)

L-A sig: adj p< 2e-02 logFC = -0.43



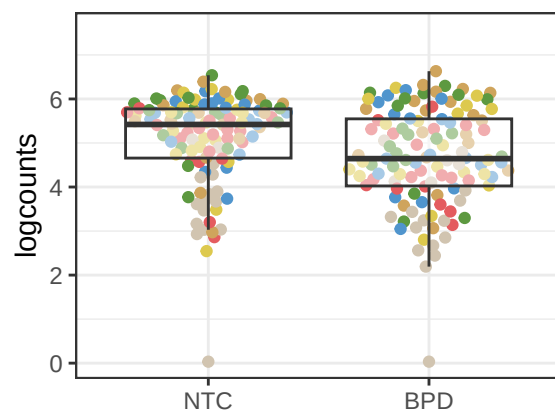
Seurat labels

L-A sig: adj p< 3e-02 logFC = -0.4

**SLC32A1**

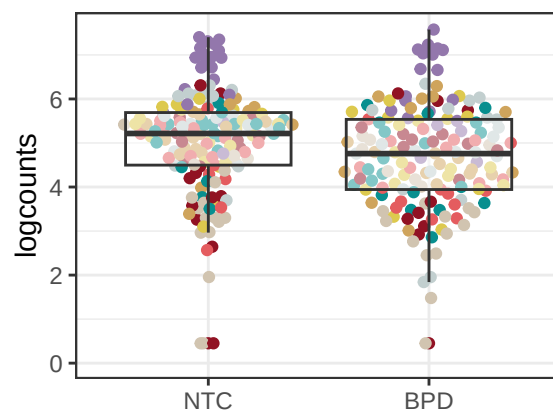
PRECAST (smoothed)

L-A sig: adj p< 2e-02 logFC = -0.49



Seurat labels

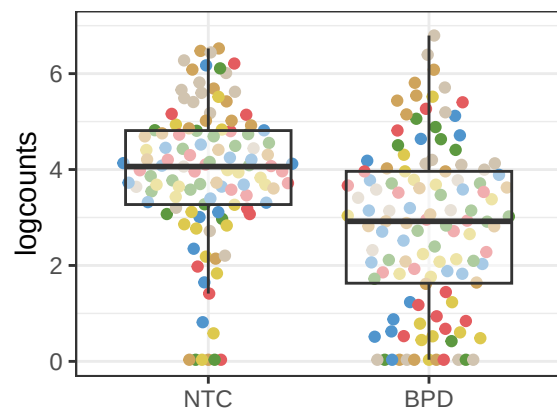
L-A sig: adj p< 3e-02 logFC = -0.49



CX3CR1

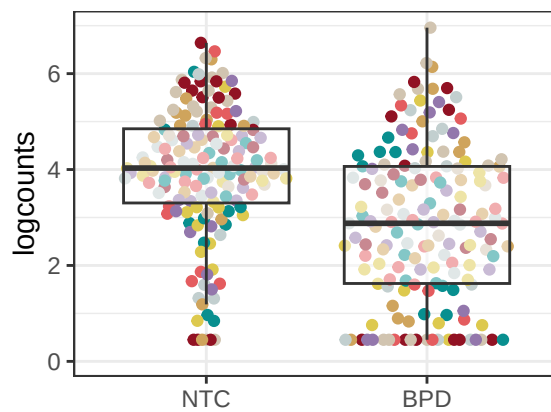
PRECAST (smoothed)

L-A sig: adj p < 1e-07 logFC = -1.52



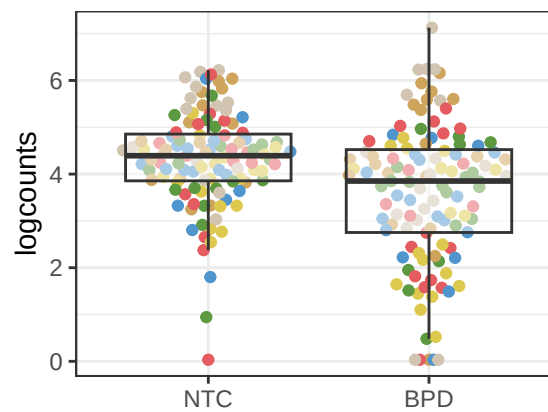
Seurat labels

L-A sig: adj p < 2e-08 logFC = -1.47

**CSF1R**

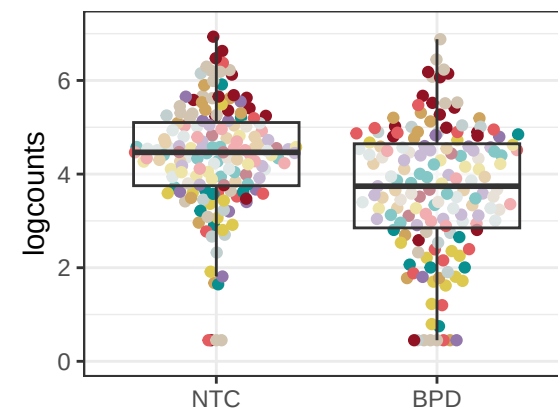
PRECAST (smoothed)

L-A sig: adj p < 1e-04 logFC = -0.93



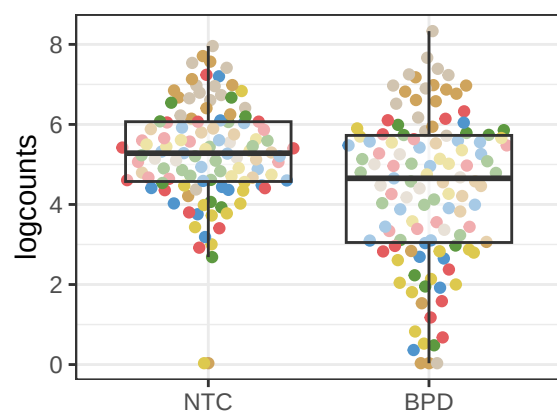
Seurat labels

L-A sig: adj p < 3e-05 logFC = -0.9

**C3**

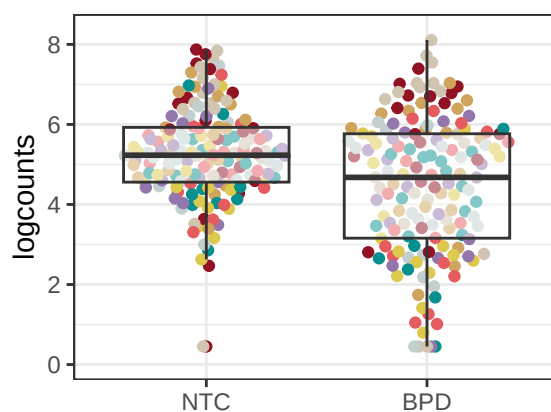
PRECAST (smoothed)

L-A sig: adj p < 3e-04 logFC = -0.97



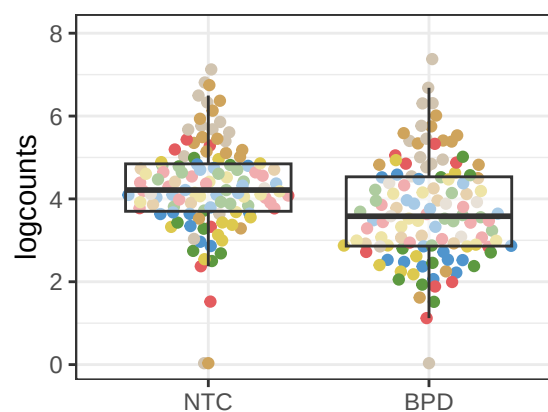
Seurat labels

L-A sig: adj p < 2e-05 logFC = -1

**HLA-DPA1**

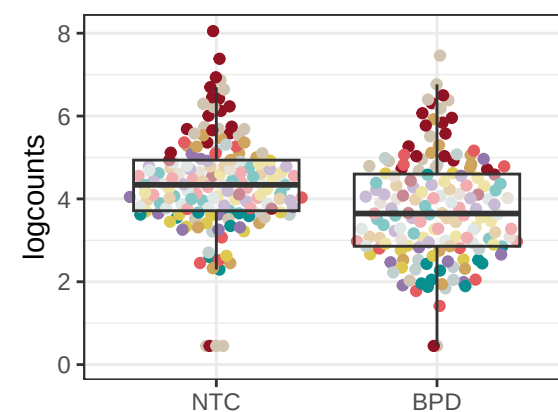
PRECAST (smoothed)

L-A sig: adj p < 6e-04 logFC = -0.76



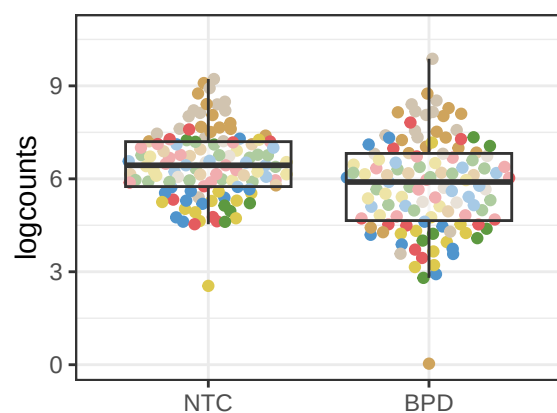
Seurat labels

L-A sig: adj p < 2e-04 logFC = -0.73

**CD74**

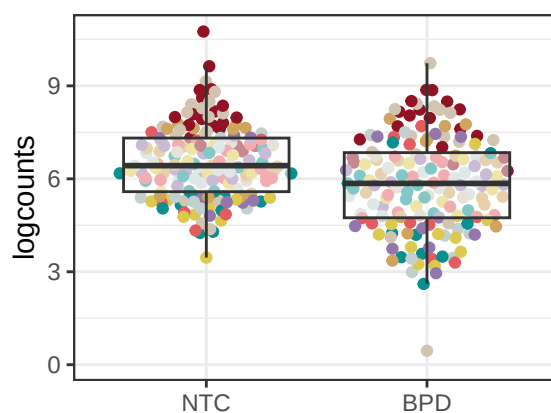
PRECAST (smoothed)

L-A sig: adj p < 4e-03 logFC = -0.65



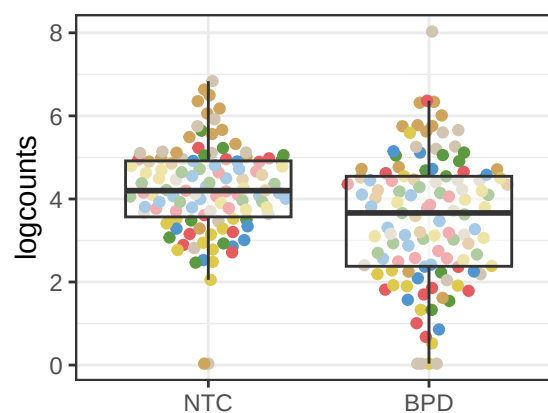
Seurat labels

L-A sig: adj p < 5e-04 logFC = -0.64

**C1QA**

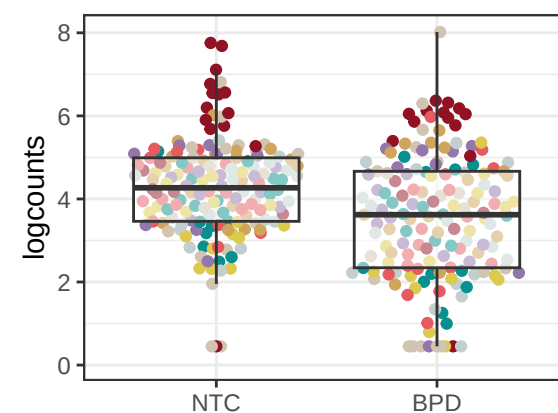
PRECAST (smoothed)

L-A sig: adj p < 1e-02 logFC = -0.74



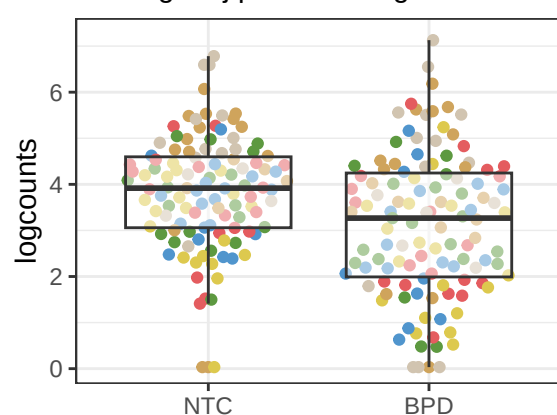
Seurat labels

L-A sig: adj p < 4e-03 logFC = -0.74

**TYROBP**

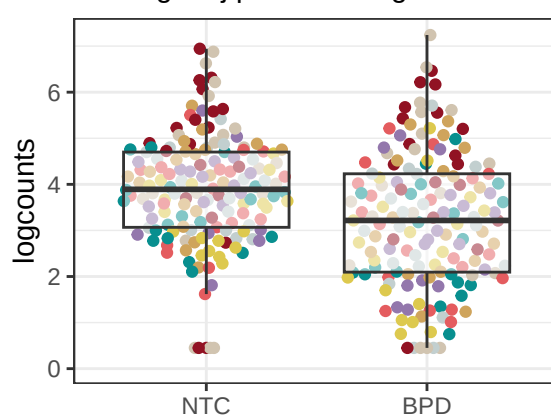
PRECAST (smoothed)

L-A sig: adj p < 1e-02 logFC = -0.73



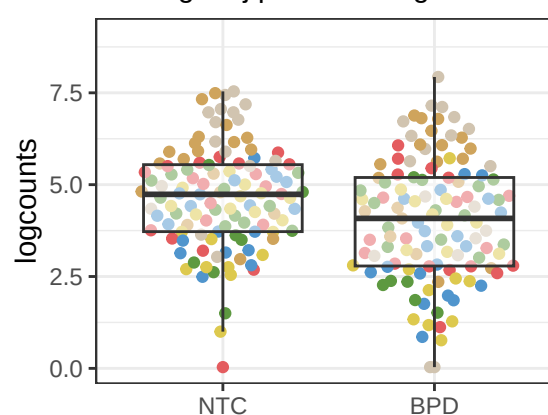
Seurat labels

L-A sig: adj p < 2e-03 logFC = -0.79

**HLA-DRA**

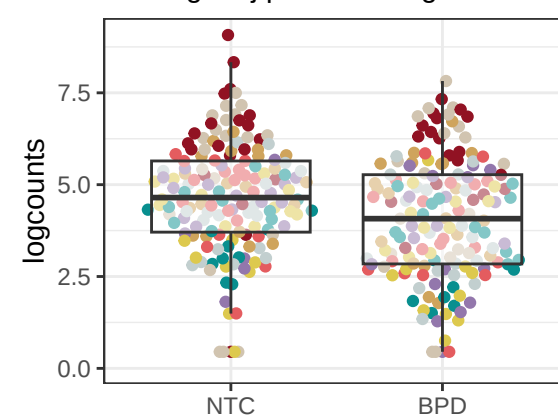
PRECAST (smoothed)

L-A sig: adj p < 1e-02 logFC = -0.72



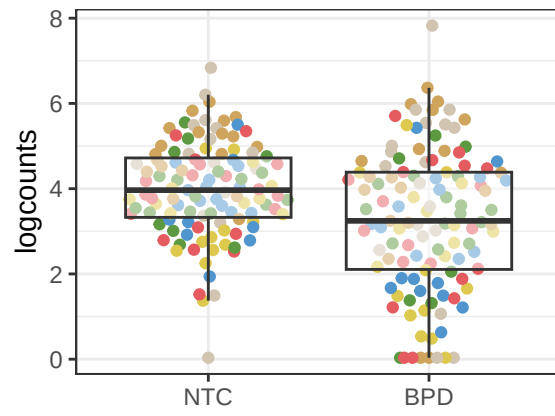
Seurat labels

L-A sig: adj p < 4e-03 logFC = -0.74

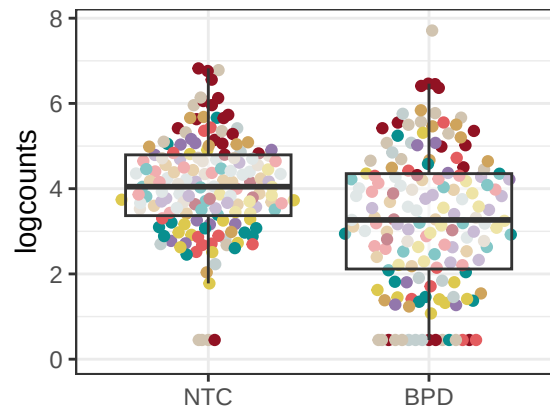


AIF1**PRECAST (smoothed)**

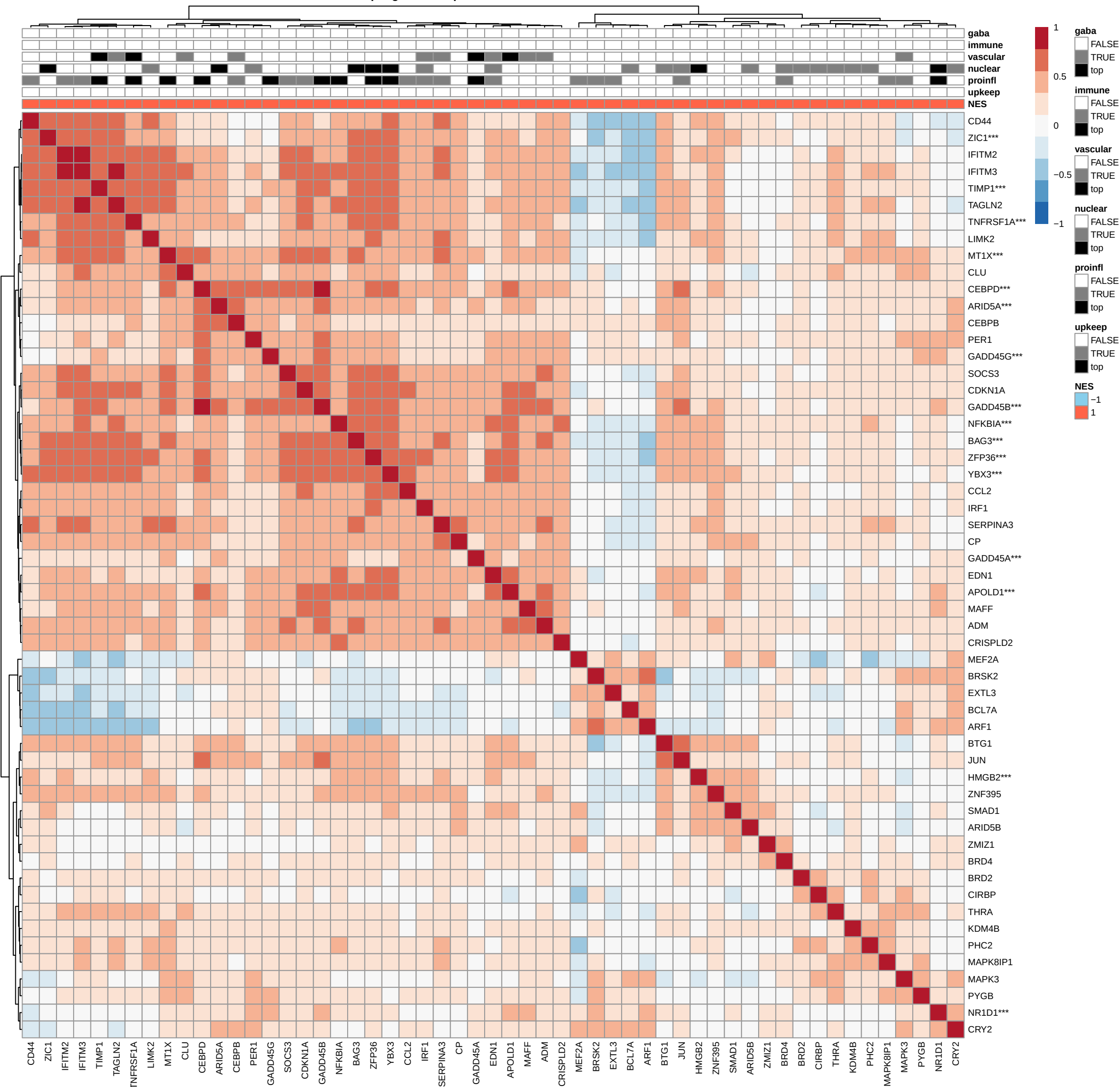
L-A sig: adj p < 2e-02 logFC = -0.75

**Seurat labels**

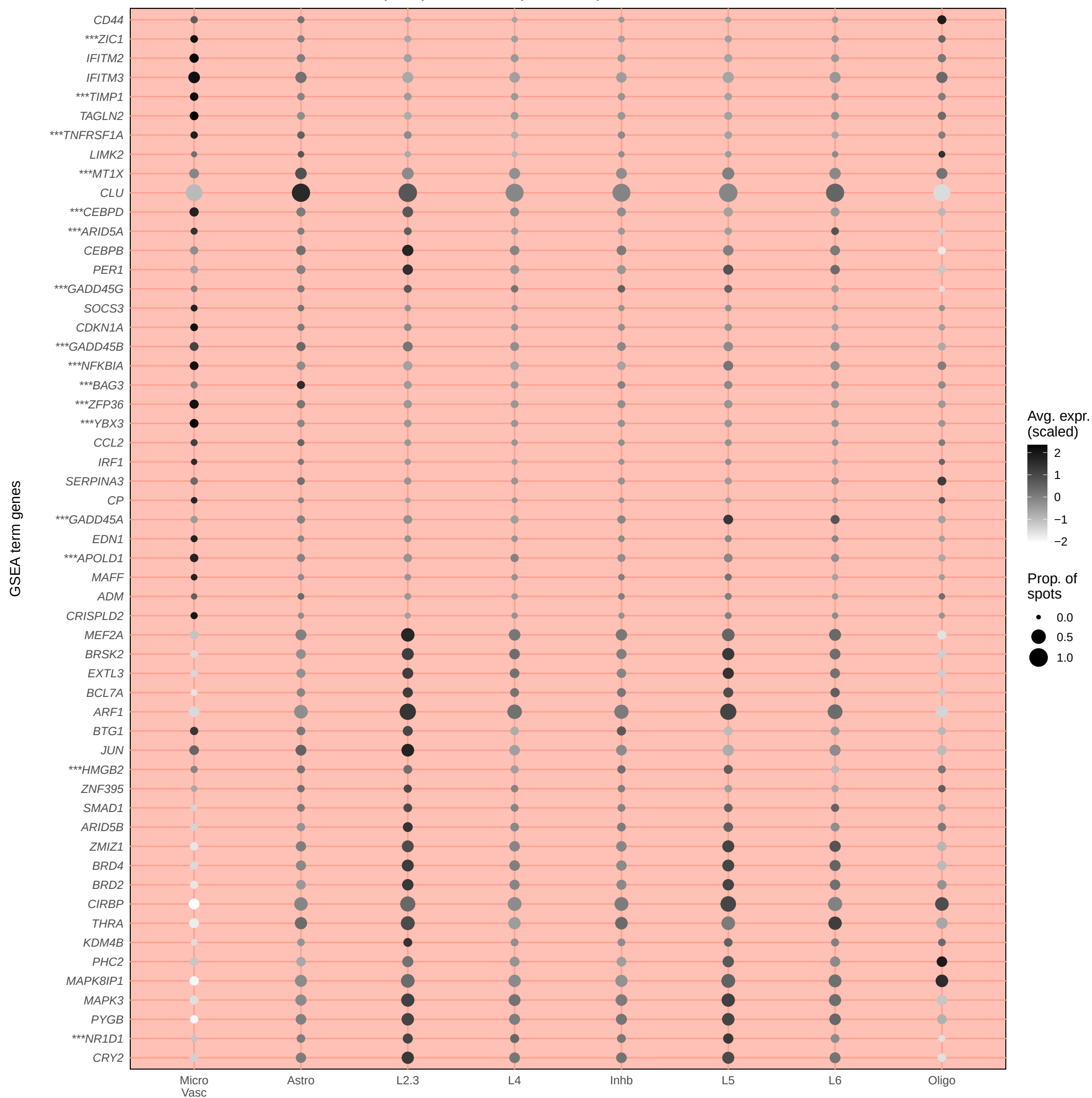
L-A sig: adj p < 2e-02 logFC = -0.69



NTC.BPD M L-A enriched genes
Top logFC DEGs per theme annotated with black



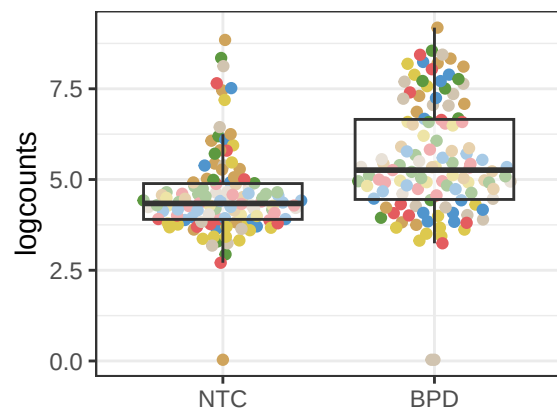
NTC.BPD M L-A enriched terms (**top L-A DEGs per theme)



GADD45B

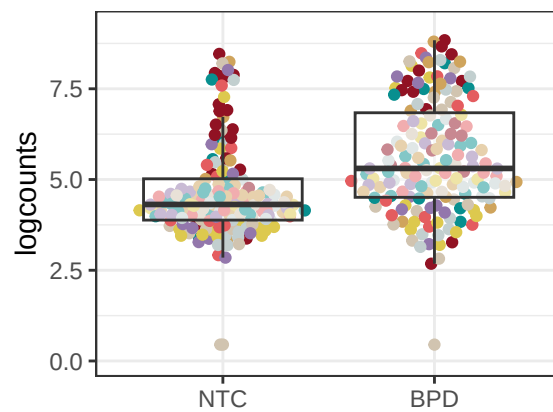
PRECAST (smoothed)

L-A sig: adj p< 2e-05 logFC = 1.16



Seurat labels

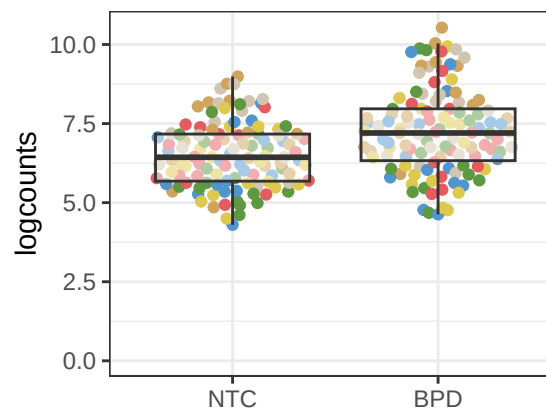
L-A sig: adj p< 1e-07 logFC = 1.21



MT1X

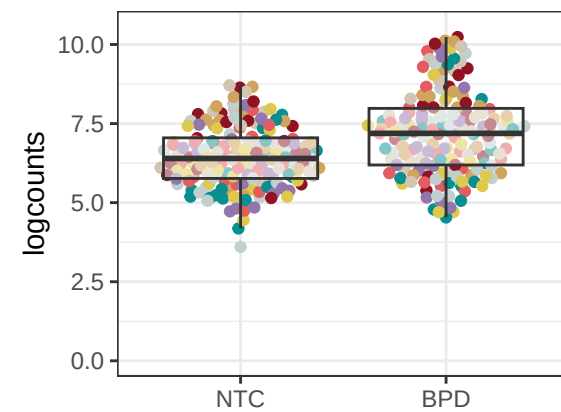
PRECAST (smoothed)

L-A sig: adj p< 3e-04 logFC = 0.82



Seurat labels

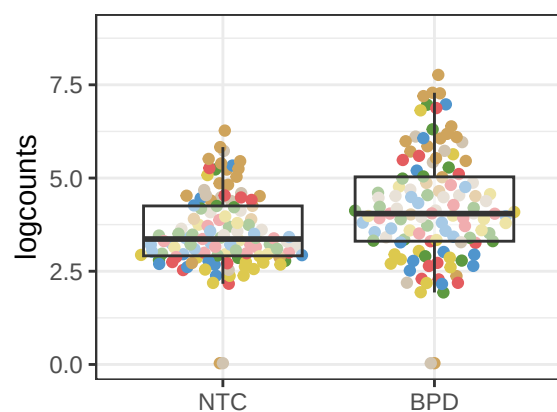
L-A sig: adj p< 2e-05 logFC = 0.83



YBX3

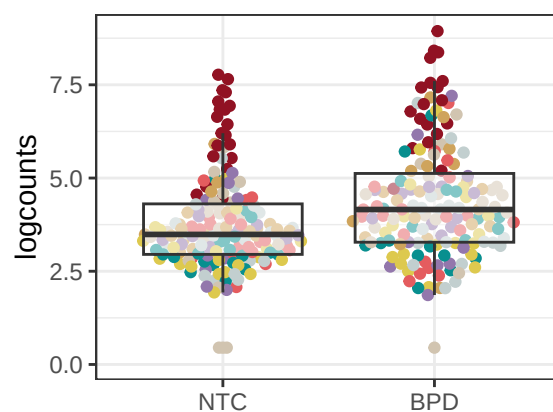
PRECAST (smoothed)

L-A sig: adj p< 1e-03 logFC = 0.81



Seurat labels

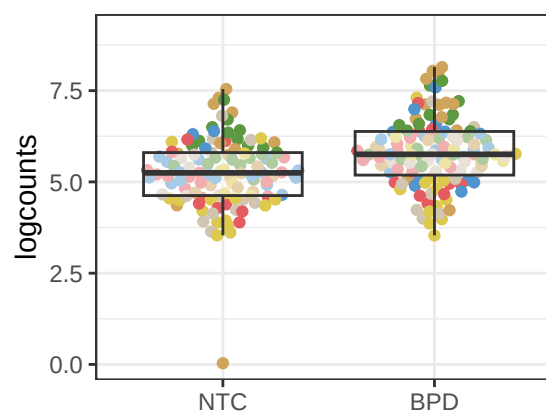
L-A sig: adj p< 4e-04 logFC = 0.79



CEBPD

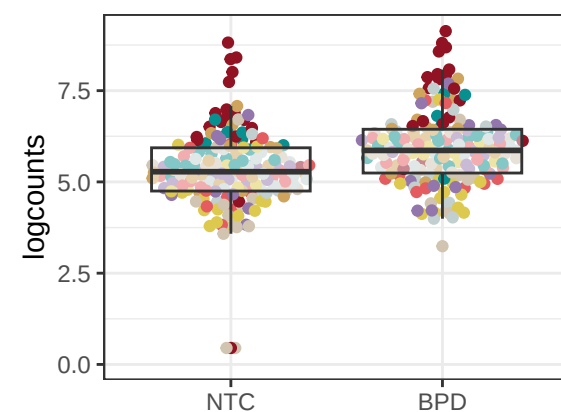
PRECAST (smoothed)

L-A sig: adj p< 1e-03 logFC = 0.58



Seurat labels

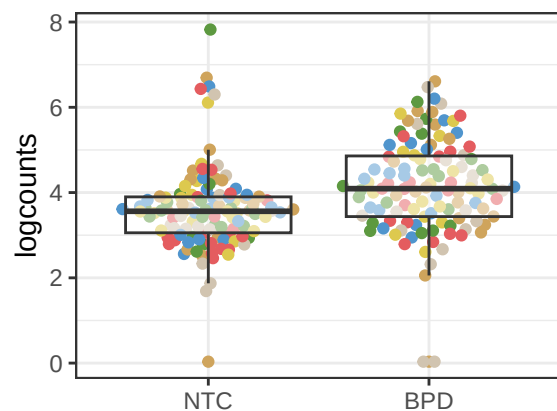
L-A sig: adj p< 1e-04 logFC = 0.59



GADD45G

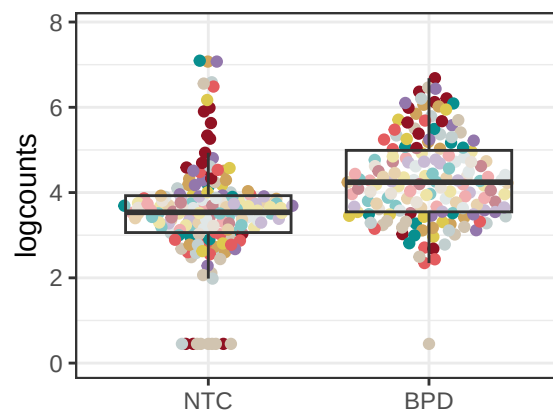
PRECAST (smoothed)

L-A sig: adj p< 1e-03 logFC = 0.74



Seurat labels

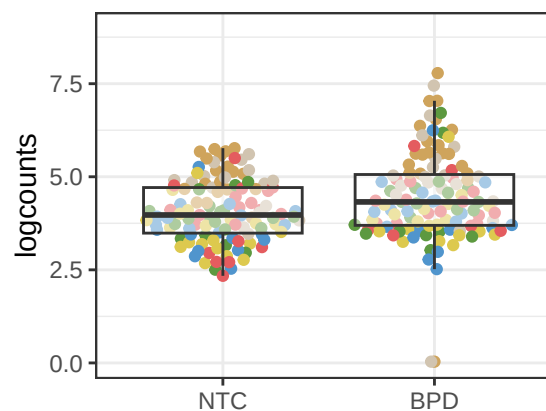
L-A sig: adj p< 3e-03 logFC = 0.66



TIMP1

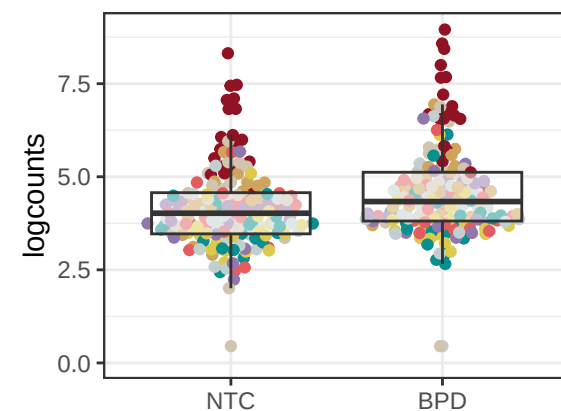
PRECAST (smoothed)

L-A sig: adj p< 4e-03 logFC = 0.58



Seurat labels

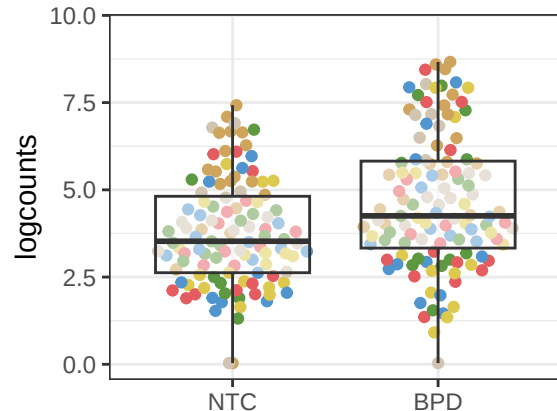
L-A sig: adj p< 5e-03 logFC = 0.55



ZFP36

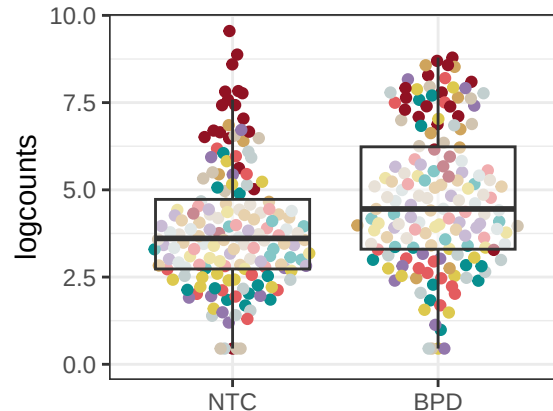
PRECAST (smoothed)

L-A sig: adj p< 1e-02 logFC = 1



Seurat labels

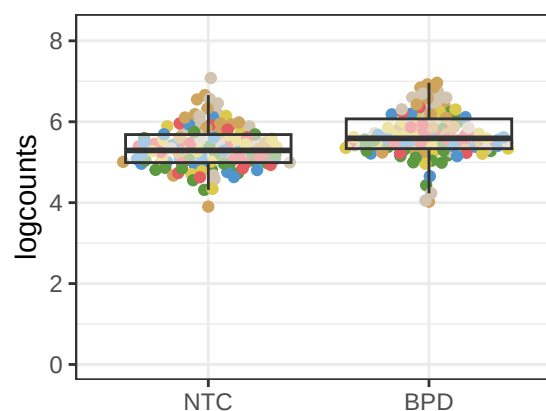
L-A sig: adj p< 9e-03 logFC = 0.91



NFKBIA

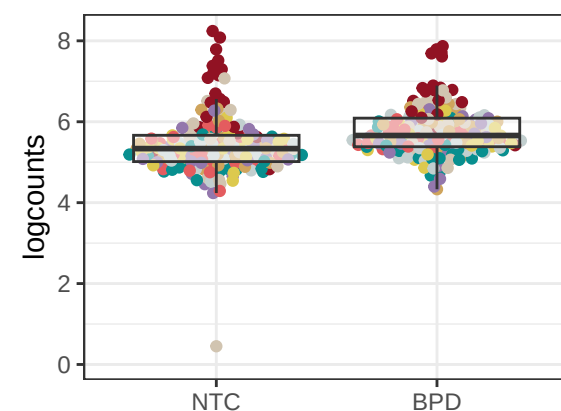
PRECAST (smoothed)

L-A sig: adj p< 1e-02 logFC = 0.33



Seurat labels

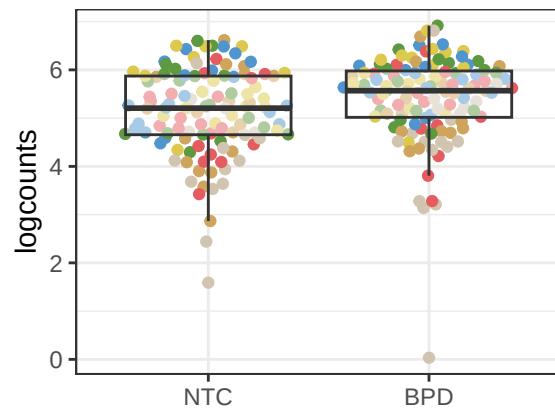
L-A sig: adj p< 1e-02 logFC = 0.32



NR1D1

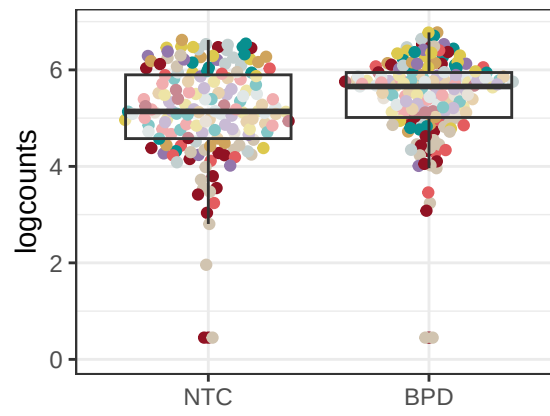
PRECAST (smoothed)

L-A sig: adj p< 1e-02 logFC = 0.39



Seurat labels

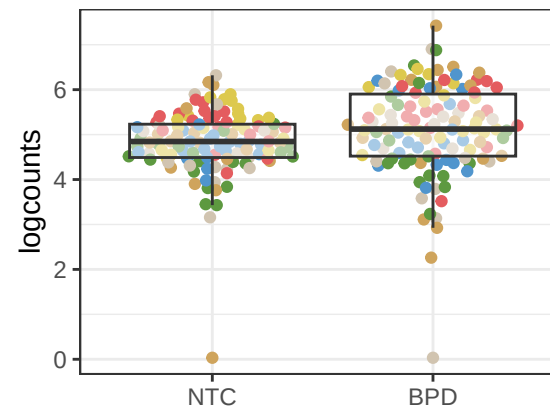
L-A sig: adj p< 1e-02 logFC = 0.38



GADD45A

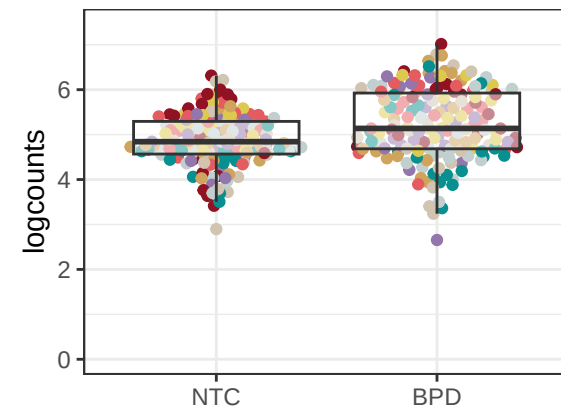
PRECAST (smoothed)

L-A sig: adj p< 4e-02 logFC = 0.42



Seurat labels

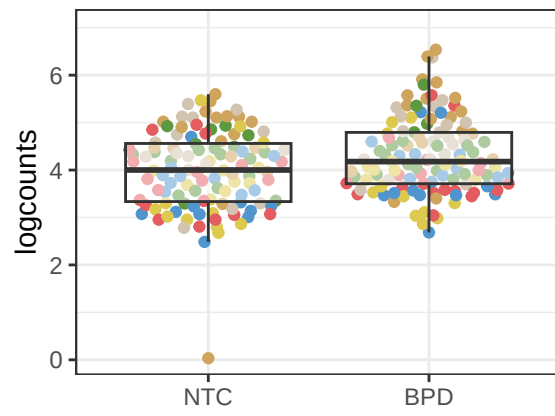
L-A sig: adj p< 2e-02 logFC = 0.44



TNFRSF1A

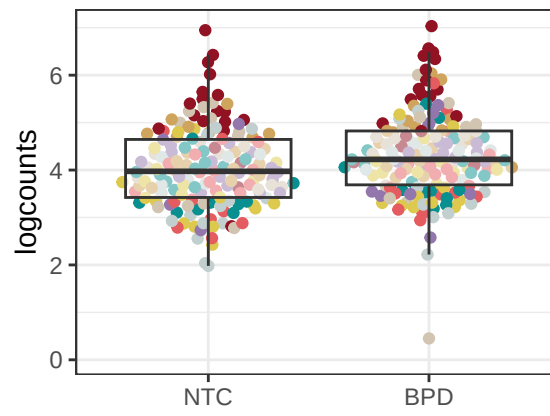
PRECAST (smoothed)

L-A sig: adj p< 4e-02 logFC = 0.4



Seurat labels

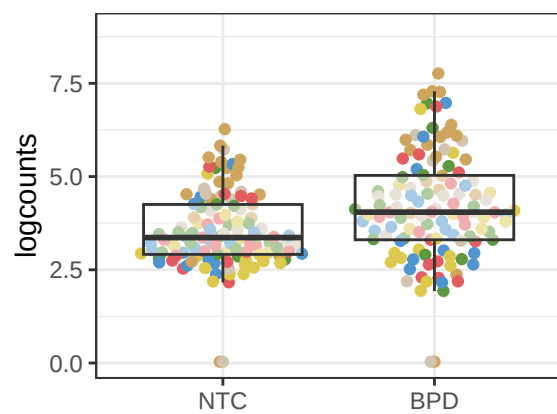
L-A sig: adj p< 2e-02 logFC = 0.42



YBX3

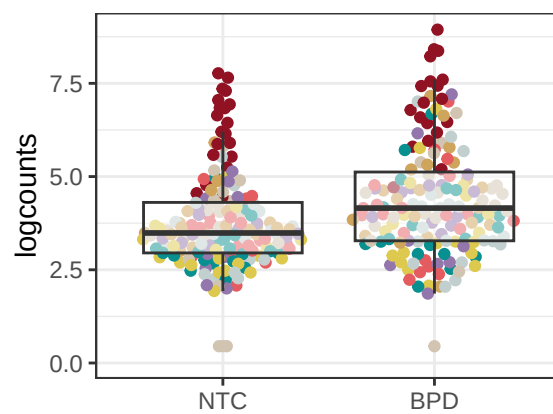
PRECAST (smoothed)

L-A sig: adj p< 1e-03 logFC = 0.81



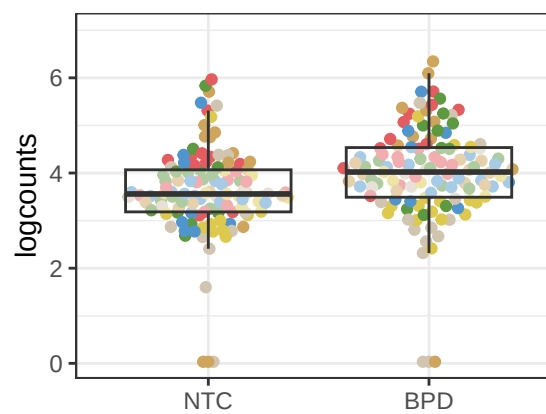
Seurat labels

L-A sig: adj p< 4e-04 logFC = 0.79

**ARID5A**

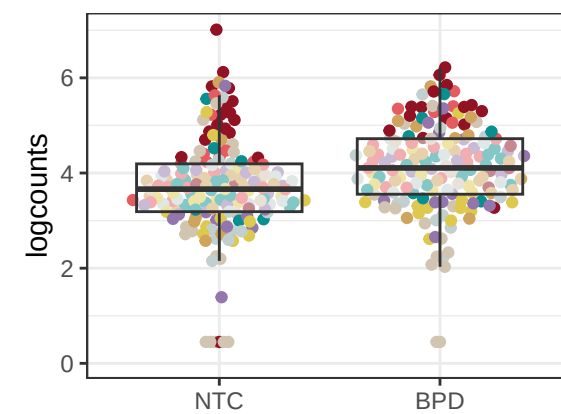
PRECAST (smoothed)

L-A sig: adj p< 2e-03 logFC = 0.55



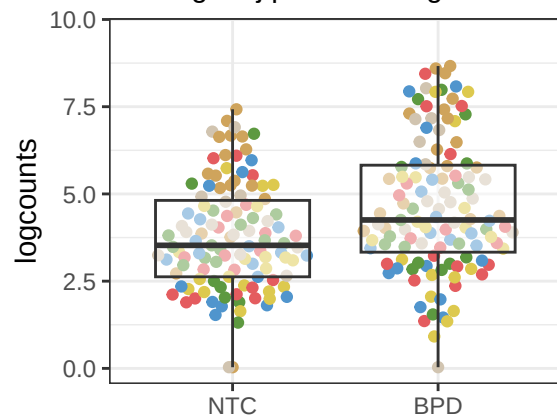
Seurat labels

L-A sig: adj p< 8e-04 logFC = 0.57

**ZFP36**

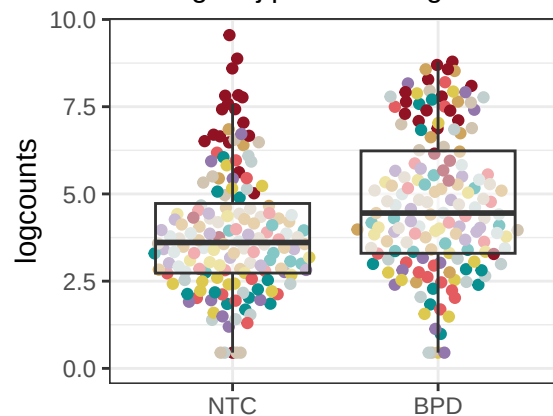
PRECAST (smoothed)

L-A sig: adj p< 1e-02 logFC = 1



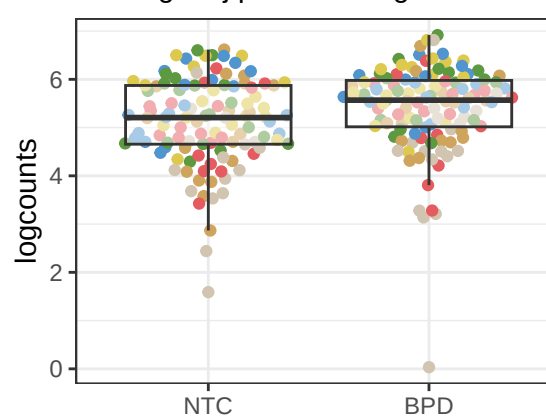
Seurat labels

L-A sig: adj p< 9e-03 logFC = 0.91

**NR1D1**

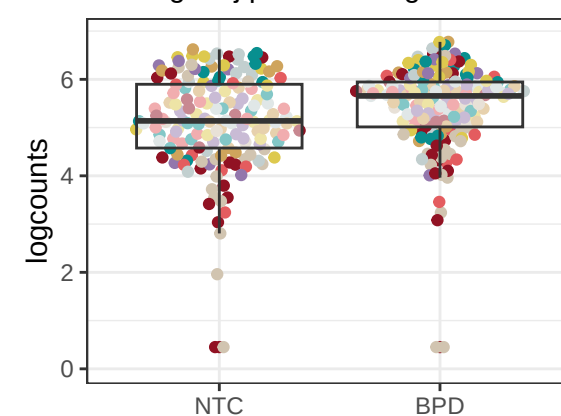
PRECAST (smoothed)

L-A sig: adj p< 1e-02 logFC = 0.39



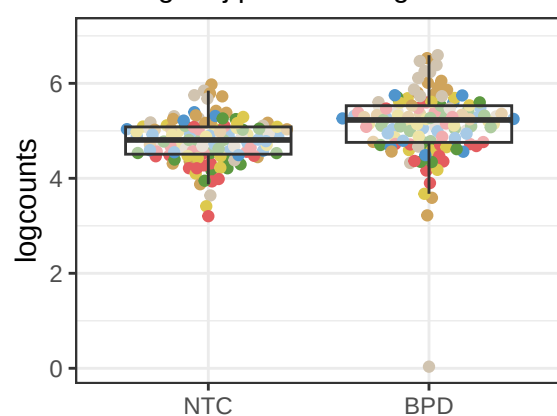
Seurat labels

L-A sig: adj p< 1e-02 logFC = 0.38

**HMGB2**

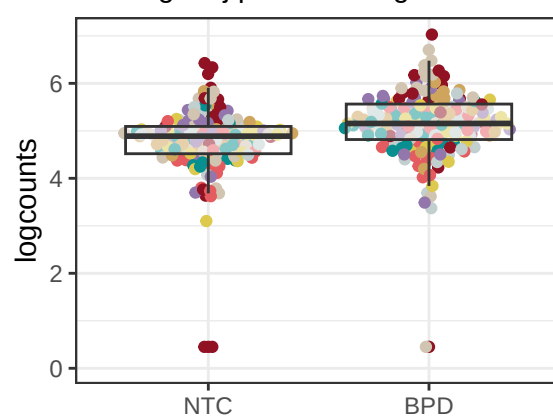
PRECAST (smoothed)

L-A sig: adj p< 2e-02 logFC = 0.3



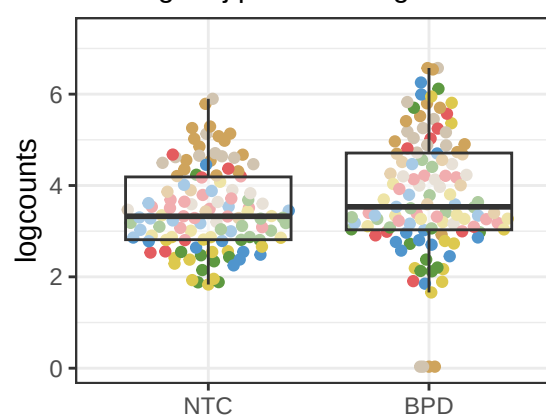
Seurat labels

L-A sig: adj p< 2e-02 logFC = 0.31

**ZIC1**

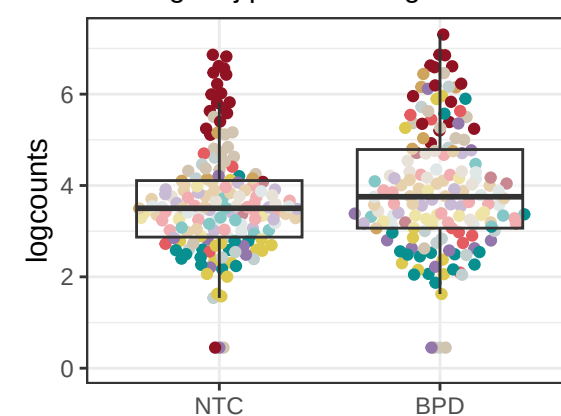
PRECAST (smoothed)

L-A sig: adj p< 4e-02 logFC = 0.51



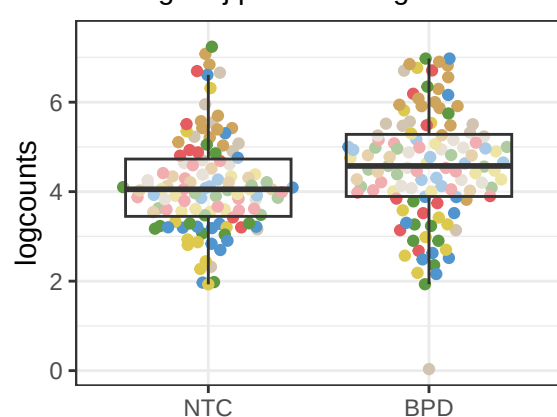
Seurat labels

L-A sig: adj p< 3e-02 logFC = 0.53

**BAG3**

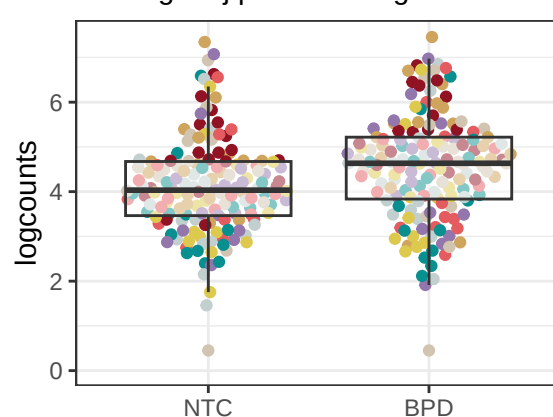
PRECAST (smoothed)

L-A sig: adj p< 4e-02 logFC = 0.6



Seurat labels

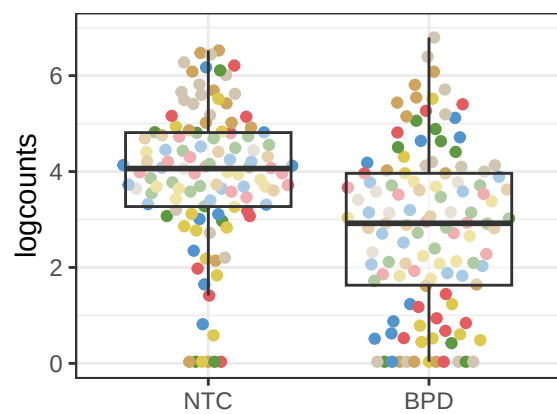
L-A sig: adj p< 1e-02 logFC = 0.65



CX3CR1

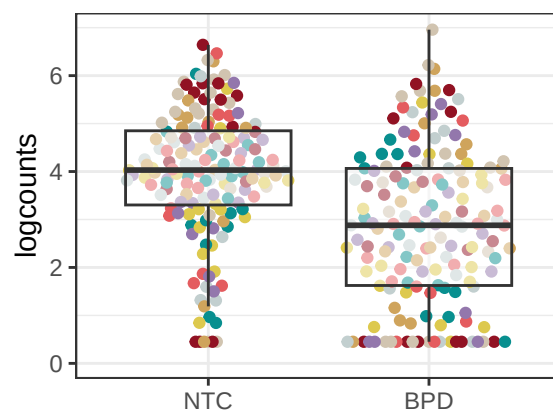
PRECAST (smoothed)

L-A sig: adj p< 1e-07 logFC = -1.52



Seurat labels

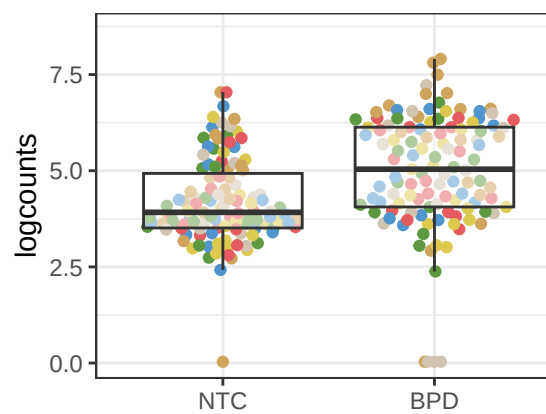
L-A sig: adj p< 2e-08 logFC = -1.47



APOLD1

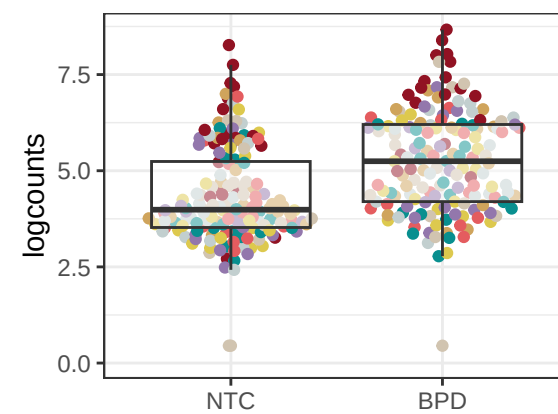
PRECAST (smoothed)

L-A sig: adj p< 2e-05 logFC = 0.95



Seurat labels

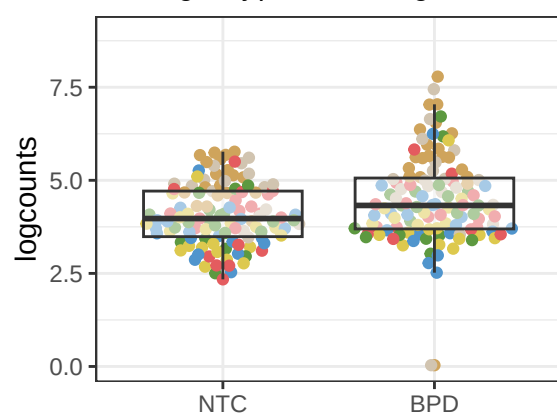
L-A sig: adj p< 2e-06 logFC = 0.9



TIMP1

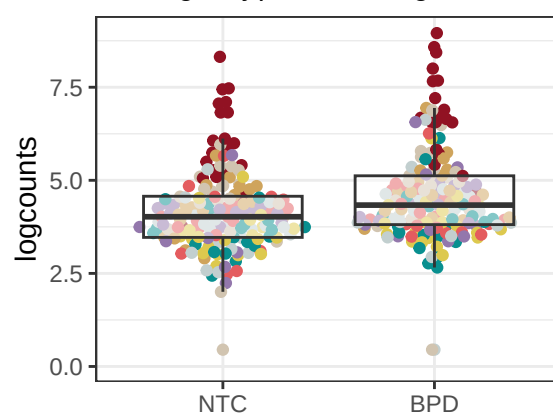
PRECAST (smoothed)

L-A sig: adj p< 4e-03 logFC = 0.58



Seurat labels

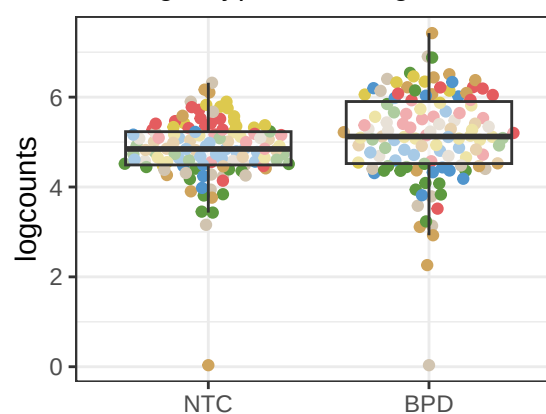
L-A sig: adj p< 5e-03 logFC = 0.55



GADD45A

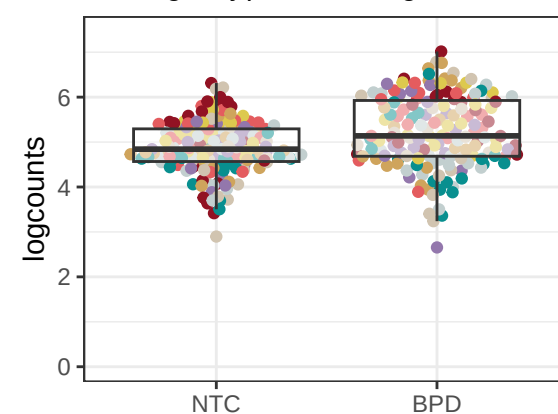
PRECAST (smoothed)

L-A sig: adj p< 4e-02 logFC = 0.42



Seurat labels

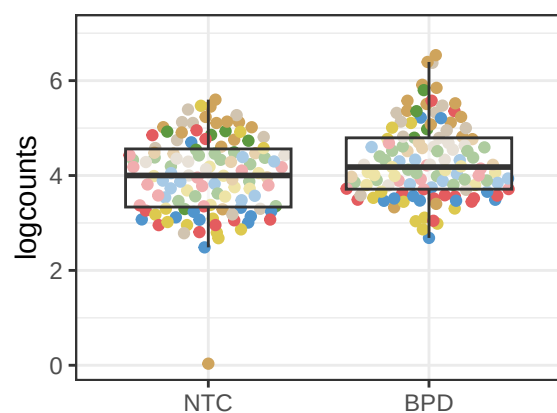
L-A sig: adj p< 2e-02 logFC = 0.44



TNFRSF1A

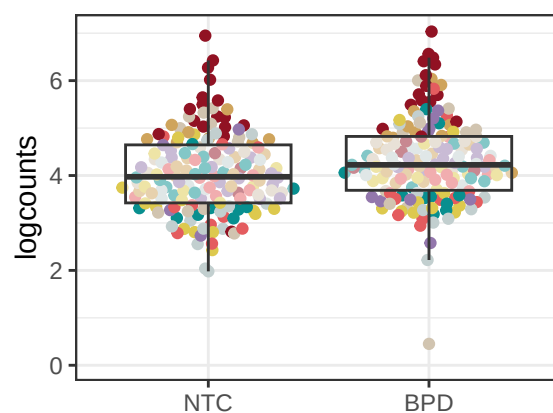
PRECAST (smoothed)

L-A sig: adj p< 4e-02 logFC = 0.4

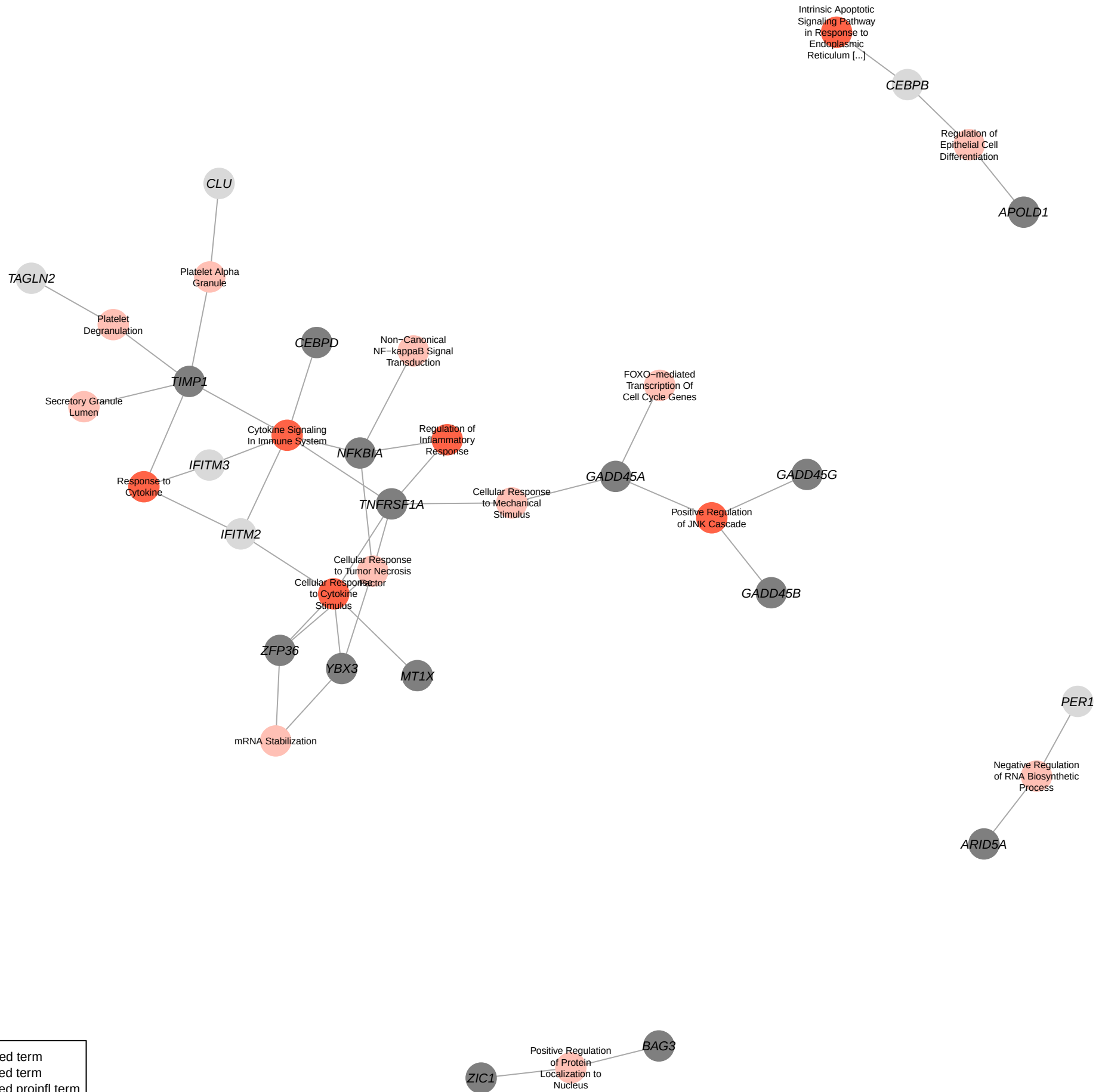


Seurat labels

L-A sig: adj p< 2e-02 logFC = 0.42



Reactome, GO-BP, and GO-CC results combined to produce theme



NTC.BPD M enriched (neuronal)

